Geometric Sequences and Series

ANSWER KEY

Section 1: General Term of a Geometric Sequence

1 State the formula for the n-th term a_n of a geometric sequence with first term a_1 and common ratio r.

a. $a_n = a_1 r^{n-1}$ ✓

b. $a_n = a_1 r^n$

c. $a_n = a_1 + (n-1)r$

d. $a_n = \frac{a_1}{r^{n-1}}$

In a geometric sequence, each successive term is multiplied by r, giving $a_n = a_1 r^{n-1}$.

2 Find the 6th term of the geometric sequence $3, 6, 12, 24, \ldots$

a. 96 ✓

b. 192

c. 48

d. 128

$a_1 = 3, r = 2$. $a_6 = 3 \cdot 2^{6-1} = 3 \cdot 2^5 = 3 \cdot 32 = 96$.

3 In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio r.

a. 3 ✓

b. 2

c. 4

d. $\sqrt{3}$

$\frac{a_5}{a_2} = r^3 = \frac{324}{12} = 27 \Rightarrow r = 3$.

Section 2: Finite Geometric Series

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4 State the formula for the sum S_n of the first n terms of a geometric series ($r \neq 1$).

a. $S_n = \frac{a_1(1-r^n)}{1-r}$ ✓

b. $S_n = \frac{a_1}{1-r}$

c. $S_n = \frac{n}{2}(a_1 + a_n)$

d. $S_n = a_1(1-r^n)$

The finite sum of a geometric series is $S_n = \frac{a_1(1-r^n)}{1-r}$.

5 Find the sum of the first 6 terms of the series $4 + 12 + 36 + \ldots$

a. 1456 ✓

b. 1458

c. 486

d. 2916

$a_1 = 4, r = 3, n = 6$. $S_6 = \frac{4(1-3^6)}{1-3} = \frac{4(1-729)}{-2} = -2(-728) = 1456$.

6 Evaluate $\sum_{k=1}^{5} 2(3)^{k-1}$.

a. 242 ✓

b. 162

c. 243

d. 486

$a_1 = 2, r = 3, n = 5$. $S_5 = \frac{2(1-3^5)}{1-3} = \frac{2(1-243)}{-2} = 242$.

Section 3: Infinite Geometric Series

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7 Under what condition does an infinite geometric series $\sum_{n=1}^{\infty} a_1 r^{n-1}$ converge?

a. $|r| < 1$ ✓

b. $r > 1$

c. $|r| \leq 1$

d. $r > 0$

An infinite geometric series converges if and only if the absolute value of the common ratio is strictly less than 1 ($|r| < 1$).

8 Find the sum of the infinite series $12 + 4 + \frac{4}{3} + \frac{4}{9} + \ldots$

a. 18 ✓

b. 16

c. 24

d. 15

$a_1 = 12, r = 1/3$. Since $|1/3| < 1$, sum $S = \frac{a_1}{1-r} = \frac{12}{1-1/3} = \frac{12}{2/3} = 18$.

9 Find the fractional equivalent of the repeating decimal $0.\overline{45} = 0.454545\ldots$ using an infinite geometric series.

a. 5/11 ✓

b. 45/100

c. 9/22

d. 45/90

$a_1 = 45/100, r = 1/100$. $S = \frac{45/100}{1-1/100} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}$.

Section 4: Geometric Word Problems and Annuities

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10 A bouncing ball is dropped from a height of 16 ft. On each bounce, it rebounds to $3/4$ of its previous height. Find the total vertical distance the ball travels before coming to rest.

a. 112 ft ✓

b. 64 ft

c. 128 ft

d. 96 ft

Initial drop $= 16$. Subsequent up-and-down trips form an infinite series with first term $2(16 \times 3/4) = 24$ and ratio $3/4$. Total $= 16 + \frac{24}{1 - 3/4} = 16 + \frac{24}{1/4} = 16 + 96 = 112$ ft.

11 A company offers a starting salary of 50000 dollars with a guaranteed 4% annual raise. What is the salary in the 5th year?

a. $50000(1.04)^4 \approx 58492.93$ dollars ✓

b. 60000 dollars

c. 58000 dollars

d. $50000(1.04)^5$

5th year salary is $a_5 = a_1 r^4 = 50000(1.04)^4 \approx 58492.93$ dollars.

12 An ordinary annuity pays 100 dollars at the end of each year for 10 years at 5% annual interest. Which expression represents its future value?

a. $100\left[\frac{(1.05)^{10} - 1}{0.05}\right]$ ✓

b. $100(1.05)^{10}$

c. $1000(1.05)$

d. $\frac{100}{0.05}$

The sum of an annuity is a finite geometric series with formula $FV = P\left[\frac{(1+r)^n - 1}{r}\right] = 100\left[\frac{(1.05)^{10} - 1}{0.05}\right]$.

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